STEPHANIE KRIEWALL

Data Science | Machine Learning

206-819-1846 | slkriewall@gmail.com | Thornton, CO
linkedin.com/in/skriewall | github.com/skriewall | skriewall.github.io

PROFESSIONAL SUMMARY

Master's in Statistics with strong programming and statistical analysis skills, and hands-on experience applying data science and machine learning techniques through academic and independent projects. Proven ability to adapt to new domains and translate complex data into actionable insights for data-driven decision-making.

SKILLS

Programming: Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow/Keras, PySpark), SQL, R, C++, SAS, JavaScript

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Association Rule Learning, Causal Impact Analysis, Neural Networks

Tools: Excel, Tableau, Git, AWS, Anaconda, AI (Generative AI, LLMs, Agentic AI)

Math: Statistics, A/B Testing, Time Series Analysis, Bayesian Statistics, Linear Algebra

PROJECTS

Enhancing Targeting Accuracy Using Machine Learning

- Used **logistic regression**, **decision tree**, **random forest**, and **KNN** models on grocery transaction data to predict customer sign-up probabilities for a grocery delivery service based on available customer data, enabling better targeting decisions based on which customers are most likely to sign up.

"You Are What You Eat" Customer Segmentation

- Performed **k-means clustering** on grocery transaction data to split customers into distinct "shopper types" that could be used to more accurately target customers with relevant content and promotions and to better understand customers over time.

WORK EXPERIENCE

SOFTWARE DEVELOPER, Greenwood Village, CO

Mentice, May 2019-August 2025

- Developed and maintained core simulation framework for endovascular and neurovascular procedures (VIST product line), enabling hands-on training with real-time physics and medical device interaction. Used by 700+ customers – including hospitals, universities, simulation centers, and top 20 MedTech companies – to support clinical decision-making and improve patient outcomes.
- Created a 3D augmented reality surgical training tool in Unreal Engine, providing immersive visualization of heart procedures for enhanced clinician engagement and learning.
- Solved a critical Unreal Engine adaptation challenge for medical simulation by implementing the centripetal parameterization of Catmull-Rom splines, preventing cusps and self-intersections in tortuous vascular anatomy.
- Researched and developed an adaptive spline simplification algorithm to reduce control point density and vessel mesh complexity while preserving anatomical fidelity, significantly improving runtime performance in high-

complexity cases.

- Automated build workflows using Python, including a custom installer that seamlessly integrates Unreal Engine applications with the VIST simulation environment.
- Contributed to the full development lifecycle as part of a cross-functional Agile team, collaborating closely with fellow engineers and 3D designers.

MATHEMATICAL ANALYST, Wheat Ridge, CO

Gaming Laboratories International, March 2015-May 2019

- Developed C++ game simulations and used probability models to analyze the return to player and odds for slot machine games.
- Performed statistical analysis of simulations and conducted RNG randomness testing to ensure regulatory compliance.
- Evaluated player strategy and house edge for table and slot games using concepts including probability theory, expected value, Markov decision processes, dynamic programming, and game theory.
- Trained new analysts and conducted detailed technical reviews to ensure accuracy and consistency in game evaluations.

ANALYST, Greenwood Village, CO

FIMAC Solutions LLC, November 2014-March 2015

- Conducted prepayment modeling and other statistical analysis to support risk and performance assessments for financial institutions.
- Developed code for integration into company software products.

EDUCATION

MASTER OF SCIENCE IN STATISTICS, Iowa City, IA

The University of Iowa, May 2014

BACHELOR OF ARTS IN MATHEMATICS, Milwaukee, WI

Wisconsin Lutheran College, May 2010

COURSES AND CERTIFICATIONS

Data Science Professional Certification (Data Science Infinity)

Actionable Learnings: Extracted, cleaned, and manipulated data using SQL and Python. Applied statistical inference, A/B testing, and causal impact analysis. Built regression, classification, clustering, and association rules using full ML workflows (feature engineering, model validation, and automated ML pipelines). Created dashboards in Tableau and deployed a complete ML system to a live Streamlit web application. Gained practical experience in Generative AI, including transformer fundamentals, LLM behavior, prompt engineering, RAG systems, and Agentic AI workflows. Worked with AWS (S3, Lambda, CloudShell, EC2, and Sagemaker) for storage, compute, and deployment patterns. Focused throughout on turning ambiguous business problems into measurable, data-driven solutions.

Introduction to Big Data with PySpark Course (Codecademy)

Actionable Learnings: Worked with RDDs and DataFrames to transform and aggregate large datasets. Applied Spark SQL for querying and created reusable temp views.